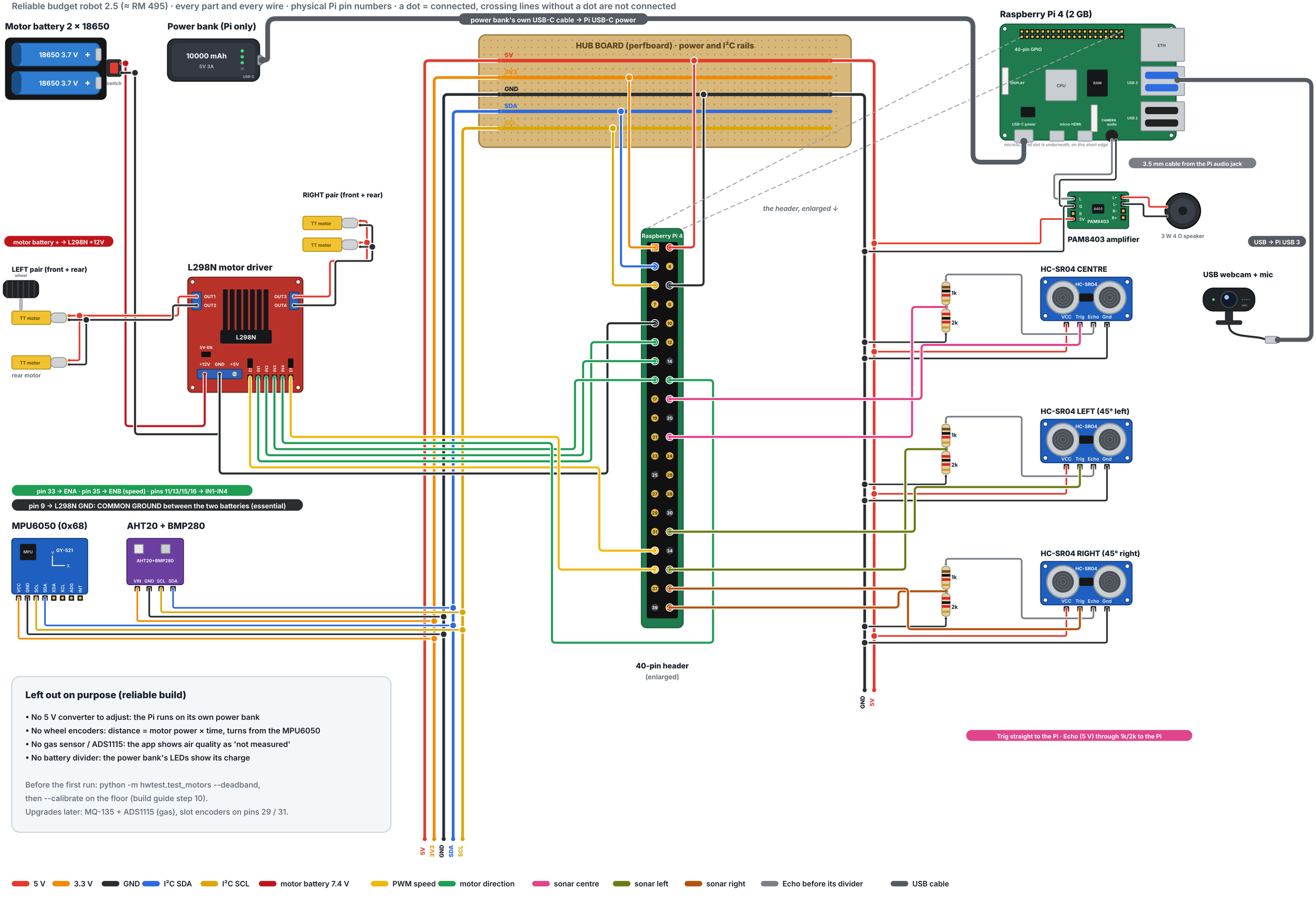
RAQIB · FULL WIRING DIAGRAM

Reliable budget robot 2.5 (≈ RM 495) · every part and every wire · physical Pi pin numbers · a dot = connected, crossing lines without a dot are not connected



Left out on purpose (reliable build)

- No 5 V converter to adjust: the Pi runs on its own power bank
- No wheel encoders: distance = motor power × time, turns from the MPU6050
- No gas sensor / ADS1115: the app shows air quality as 'not measured'
- No battery divider: the power bank's LEDs show its charge

Before the first run: python -m hwtest.test_motors --deadband,
then --calibrate on the floor (build guide step 10).
Upgrades later: MQ-135 + ADS1115 (gas), slot encoders on pins 29 / 31.

5 V · 3.3 V · GND · I²C SDA · I²C SCL · motor battery 7.4 V · PWM speed · motor direction · sonar centre · sonar left · sonar right · Echo before its divider · USB cable